
An Active Learning Workflow for Efficient X-Ray Digitization

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Abstract

Medical video imaging collects large quantities of often low variation videos that need labeling for downstream studies, yet existing tools face limitations in processing throughput, leaving millions of images unlabeled and unused in experimentation efforts. Fine-Tuning of pretrained supervised deep learning models allows researchers more data processing efficiency without the loss of the context of their research. We tested two experiments to understand the learning ability of a pre-trained key point identification deep learning model (DeepLabCut). We uncovered two key findings: (1) the large-input study identified the upper limits of input data volume and training epochs before learning plateaued for videos on one species (*Serinus Canaria*); and (2) the model-refinement study identified the minimal range of diversity-based sampling (comparisons of DINOv2 embeddings or positional displacement) required for efficient and effective model training and inference on videos from two study species (*Sus scrofa*, *Serinus canaria*). Supported by a graphical user interface, we developed an active learning protocol to track 2D fluoroscopic (x-ray) data streams, optimizing for user annotation, model training and inference. We achieved an average root mean squared error (RMSE) of 0.62 pixels less and an increase of 8.52% points within 3 pixels compared to baseline models. Advances made to this programmatic workflow significantly reduce time and improve adaptability for large-scale X-ray video processing of ranging complexities to support downstream research/medical questions.

1. Introduction

Medical imaging devices often provide a large number of grayscale images that are of low contrast relative to color images. These images are information-dense, but it is challenging and time-consuming to extract this information with current methods (Provini et al., 2023). With only a limited amount of time to spend on each image, it is often physically impossible to examine all the data that might be contained in a given image stack, so the development of tools to reduce processing time is common (Aljabri et al., 2022). Work has already been done to extract information about 3D structures from static medical image stacks from computed-tomography scans, for example, bone and organ shape (Shetty et al., 2023) and tumor identification (Kashyap et al., 2025; Hsiao et al., 2022). However, there are relatively few machine learning algorithms developed to extract information from X-ray videos (Van et al., 2024; Guerrero-Tamayo et al., 2026). These videos consist of one or more (often two) X-ray emitters and camera systems recording images simultaneously of the same subject, and are most often used to evaluate the health of a patient, for example, a barium swallow study (Zarzour et al., 2025), or a specific behavior, such as walking (Ito et al., 2017), running (Anderst et al., 2009), eating and talking (Matsuo & Palmer, 2010), and even singing (Sonninen & Hurme, 1998). In non-human studies (although see (Anderst et al., 2009)), surgical-safe metal beads are often implanted in the animals to more easily track certain hard (bone) or soft tissue structures (Brainerd et al., 2010; Gatesy et al., 2010).

Radiographic (X-ray) studies have been essential to our understanding of how internal structures within the body function (e.g., (Laurence-Chasen et al., 2020; Brainerd et al., 2010; Kirkpatrick et al., 2022)). However, the rate of discovery in these X-ray video systems has been limited by the rate of data annotation (digitization). The researcher must annotate hundreds of multi-angle X-ray videos, ranging from 5 to 10 seconds, at 200 to 1000 frames s^{-1} . Annotation most often consists of a first-pass observation to ensure the study behavior is included in each video, followed by locating the pixel of each implanted marker or tissue structure (key points) both within and between video frames. For an experienced annotator using semi-automated tracking, a single video with many instances of marker occlusion (overlap) may take a researcher an entire week to annotate.

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To overcome the challenge of annotation time, computer vision algorithms and software tools, such as XMALab (Brainerd et al., 2010; Knörlein et al., 2016), have been developed. More recently, X-ray image analysis methods that leverage machine learning and artificial intelligence (AI) have proven to be more efficient annotators when given specific training data sets (Litjens et al., 2017). These AI tools may be powerful and robust across datatypes in the medical imaging field (Aljabri et al., 2022), but a challenge remains that researchers and medical personnel may not be familiar with the necessary steps in order to use these models. Including computer scientists in collaborations is mutually beneficial (Rolnick et al., 2025) by allowing greater throughput in complex data processing, more robust model development, and often optimized methods to fit models on edge devices (Hartman et al., 2025; Shuvo et al., 2023). In this study, we address these two challenges by adapting and optimizing existing machine learning models to X-ray systems and providing a user-friendly access point to train and predict with these models. Our goal is to assess the benefits of adapting existing AI models into an approachable tool used to reduce the time spent on repetitive tasks and improve accuracy.

Fine-tuning large, pretrained models has become an effective strategy for adapting general-purpose machine learning models to specialized tasks while reducing the amount of input data required (Pan & Yang, 2010; Howard & Ruder, 2018) and computer resources (GPU time, memory, etc.) (Han et al., 2024; Lv et al., 2024). DeepLabCut (DLC) models offer a variety of backbone architectures, including the lightweight ResNet-50 (He et al., 2015), to extract visual features from an image, followed by task-specific layers that predict the locations of user-defined anatomical key-points. While these models perform well on visible-light videos, X-ray images differ in contrast and texture, making it necessary to fine-tune the models with user-specific data to accurately identify structures unique to each observation (Gu & Lee, 2024). A key component of DLC is its support of iterative model updating, allowing predictions to be corrected by users and used to further fine-tune the model.

Allowing the model to select the data from which it will learn next is a critical component of many specific machine learning tasks, as output requirements and input distributions may vary over time. This method is called active learning (Cohn et al., 1994). By employing active learning frameworks, the user becomes familiar with the data without having to spend extended hours annotating repetitive data (Monarch, 2021; Lutnick et al., 2019). Active learning leverages an informed selection of input data to direct the model to perform better across the testing set (Yang et al., 2017). Finding the optimal quantity and quality that is needed for effective model learning is often task- and budget-dependent (Hacohen et al., 2022; Cacciarelli & Ku-

lahci, 2024). To further extend the efficacy of active learning efforts, building a graphical user interface (GUI) steps users through the process, supporting those with limited machine learning experience.

We developed an active learning workflow for machine-assisted data annotation of key points in these X-ray systems. While efforts to adapt the DLC models for X-ray video studies have been tested for various species and biomechanical purposes (Laurence-Chasen et al., 2020; Kirkpatrick et al., 2022; Andries et al., 2025; Flotho et al., 2024), we still lack a non-programming workflow that directly connects the data type of these studies with the DLC models and allows for iterative model updates with diversity-focused sampling and an improved location refinement head.

In using this iterative update or “stair-step” method of video tracking, we take advantage of shifting inputs (i.e., new animal poses or locations in the video) to ensure our model is best fit for the task at hand. These efforts are designed to both reduce time spent with annotations and verify the efficacy of the data that will be fed into downstream tasks. Our domain-adaptation fine-tuning protocol offers the main advantages:

1. **Ease of model updating:** By developing a graphical user interface, one can update predicted labels, retrain models, and re-predict on videos, giving exposure to data and assurance of accuracy.
2. **Computational efficiency:** By algorithmically selecting data, one can improve sample diversity and model generalizability.
3. **Location refinement:** Post-processing of model inference using traditional computer vision and mathematical algorithms can improve point identification and tracking.

2. Related Works

2.1. Annotation Tools

Many scientific or medical studies have fixed and/or limited data sets due to regulatory, ethical, and/or privacy concerns (Gröger et al., 2025; Alzubaidi et al., 2023), thus lacking the quantity of data to create an unsupervised learning model. Works in active learning have shown how systematic selection of data can maximize learning and minimize the time a user needs to create a training data set (Monarch, 2021; Cohn et al., 1994). Labeling of the data (i.e., annotation) is currently accomplished using semi-automated or fully automated tools with intuitive interfaces (Lutnick et al., 2019). Many annotation tools have been developed for scientific and medical imaging annotation, including one for X-ray of moving morphology (XROMM) studies,

XMALab (Knörlein et al., 2016). In a similar vein to model performance and changing needs, these annotation tools require maintenance and updates to stay up to date with needs and the models they run on. In the evolution of these tools, there is the opportunity to co-evolve the algorithms or models they leverage to better coincide with current tasks or systems (Isensee et al., 2024; Rolnick et al., 2025).

2.2. Diverse Data Selection for Fine-Tuning

Training datasets may contain redundant components (i.e., videos with high frame rates that produce redundant images, as in the case of this paper) that either add minimal new information to the system or risk overfitting to certain parts of the data during training of machine learning models (Ying, 2019). Therefore, finding the subset of the data that maximizes information gain not only reduces annotation time for the user, but also reduces training time and overtraining (Cohn et al., 1994; Xie et al., 2025). Data selection methods based on classical image statistics (such as pixel intensity, color distributions, and pairwise similarity) characterize low-level visual properties but may fail to capture higher-level semantic relationships between images (Liu et al., 2007). In contrast, pretrained self-supervised vision models can provide representations that encode visual structure beyond these low-level statistics (Oquab et al., 2024). With these embeddings, data selection methods can sample diverse regions of the data distribution, thereby increasing the information gained from a limited training set (Shrivastava et al., 2016; Sener & Savarese, 2018; Axiotis et al., 2024). The quantity of task-specific data required can be further reduced by leveraging models pretrained on broad datasets and adapting them to the target task, a process known as fine-tuning (Tajbakhsh et al., 2016; Ding et al., 2023). During fine-tuning, the parameters learned during pretraining are further refined to fit the features and predictions of the informative input data (Yosinski et al., 2014).

2.3. Location Refinement

Marker or pose estimation algorithms often find the area near the target, especially if, for example, there are multiple target pixels and they overlap in some frames. Instead of training the algorithm more, many studies have opted for a two-stage detection process (Nautiyal & Deshmukh, 2026; Mohammed, 2025). The two-stage detection process consists of a regional identification and a further refinement step for specific pixel(s) to be labeled (Nautiyal & Deshmukh, 2026; Mohammed, 2025). The usefulness of a two-step refinement process must be compared with single-step methods (i.e., time vs. accuracy tradeoff), and evaluating the complexity of the input images can help inform the need for two steps (Soviany & Ionescu, 2018; Carranza-García et al., 2021). The second step, where the target is located in the region informed by the first step, can be learned through deep

neural networks (Bär et al., 2023) or approached with classical computer vision techniques (Lucas & Kanade, 1981; Nebehay & Pflugfelder, 2015). This second step may be critical because there have been errors found in the annotation process in well-tested papers and datasets (Hussain et al., 2026).

3. Materials

3.1. X-Ray Data

Prior to data collection, each animal had been implanted with metal markers (1 mm of diameter for the miniature pigs and 0.3–0.4 mm for the canaries) in both hard (bone) and soft (tissue) structures relevant to each study. The implanted metal markers show up in the X-ray video as black dots that can be tracked from frame to frame. Each set of videos for a given collection event (a trial) consists of two synchronized videos, collected at approximately right angles from each other [Supplementary 1]. Two different sets of data were used in this study, 20 years apart, to test the efficacy of the workflow on both historical and newer X-ray video data. The older data (pig) contains a small quantity of markers and repetitive movements of relatively low amplitude, whereas the newer data (canary) contains higher quantities of markers and a greater range of movements, representing both simple and complex data to model.

The older data set, a feeding miniature pig (*Sus scrofa*) dataset, which is publicly available on the XMA portal (Brainerd et al., 2010; Huffman et al., 2016; Knörlein et al., 2016), was collected in 2006 at Brown University (Brainerd et al., 2010; Menegaz et al., 2015). There are 10 markers implanted in the pig. We selected 3 recorded trials totaling approximately 2,000 frames (4,000 images) from 1 individual recorded at 250 frames s^{-1} with resolution 1024×1024 . The data was also used in integrative studies of DLC into X-ray video systems (Laurence-Chasen et al., 2020).

The newer data set, a vocalising Atlantic Canary (*Serinus canaria*) data set, was collected in 2025 at the MECADEV research unit (UMR 7179). We selected 7 data collection events of 3 individuals, recorded at 750 frames s^{-1} (individual 1) and 500 frames s^{-1} (individuals 2 and 3), for a total of approximately 12,000 usable frames at a resolution of 768×768 pixels. There are 24, 20, and 24 markers implanted into Canary 1, 2, and 3, respectively.

3.2. Computer Specifications

All models are trained on a Dell Precision 3680 computer with an Intel(R) Core(TM) i7-14700 processor, 2100 MHz, 20 cores, and 28 logical processors, running Windows 11 with one NVIDIA GeForce RTX 4060 GPU.

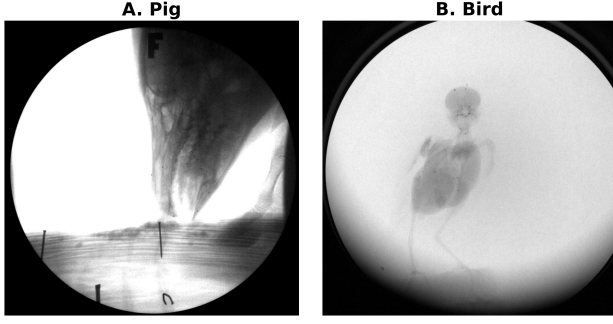


Figure 1. Raw input frames of the 2D fluoroscopic (X-ray) systems, capturing (A) the snout of Pig 1 eating food and (B) Bird 2 standing on a wooden platform. Both images were taken by Camera 1 of the study. The pig study has a higher resolution (1024×1024), more artifacts in the frame, and was recorded at $250 \text{ frames s}^{-1}$, 80–83 kV, and 12.0–12.5 mA. Bird 2 trials have a resolution of 768×768 , contain only a small wooden beam in the frame, and were recorded at $500 \text{ frames s}^{-1}$, 50 kV, and 140 mA.

4. Methods

We developed an active learning protocol (Figure 2) to track 2D fluoroscopic (X-ray) data. Each step of the new protocol was tested across input data and hyperparameter choices. Data input diversity was tested using a random sampling baseline and two diversity-focused sampling methods. Model training was tested across various input sizes and training iterations. Location refinement as a post-processing step was added to all models built under the diversity tests. We found that the optimal workflow, both for user time and training time, integrates DINOv2 (Oquab et al., 2024), DeepLabCut (Mathis et al., 2018), and an adapted difference-of-Gaussians blob detection and Hungarian algorithm for location refinement in post-processing (Lowe, 2004; Kuhn, 1955). These steps allow for optimal data selection, lightweight supervised learning, and final corrections that allow pixel-to-pixel accuracy. The framework capitalizes on data input diversity and post-prediction corrections, all controlled through a new graphical user interface (GUI) to encourage rapid improvements and limit the complexities of hyperparameter tuning that can alter the efficacy of the model.

4.1. Data Input Diversity

We tested the value of sampling a more diverse and informative subset of images for training (Sener & Savarese, 2018; Xia et al., 2024) using random and displacement-based sampling derived from the previous logic of internal DLC (Mathis et al., 2018), and information dense DINOv2 (Oquab et al., 2024) image embeddings inspired by foundational reinforcement learning, DIAYN (Eysenbach et al., 2018), and decomposed difficulty-based sampling (Ding et al., 2025).

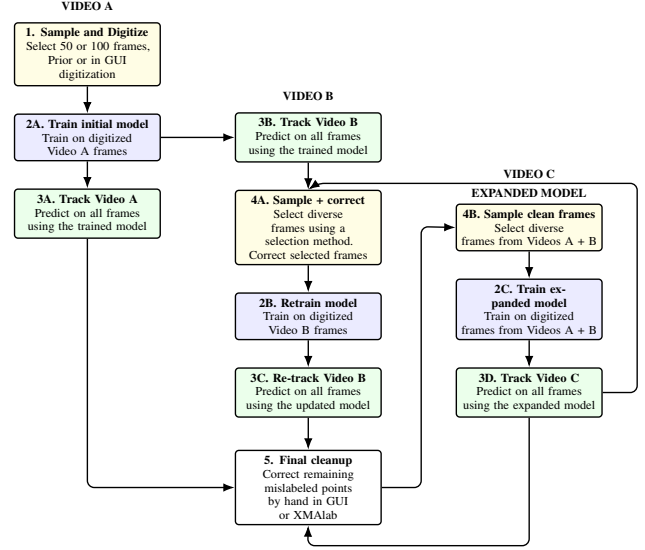


Figure 2. Iterative workflow for X-ray video digitization, model training, tracking, and refinement. Yellow boxes indicate frame selection, using DINO, Displacement, or Random, and manual correction supported by blob detection algorithm to find nearest blob to your cursor click. Blue boxes indicate model training or retraining for 125 epochs. Green boxes indicate model-based tracking and blob detection location refinement. The white boxes indicate data or final manual cleanup. The workflow begins with sampling and digitizing frames from an initial video, followed by training and tracking. Predictions from subsequent videos are selectively corrected and incorporated into the training set, allowing the model to be progressively expanded and applied to additional videos.

Random sampling was used as the baseline model and was implemented through the standard DLC dataset-creation procedure. The two other methods algorithmically target a task-specific, diverse set of input data in contrast to the conventional random-sampling approach. The first alternative, displacement sampling, is an adaptation of the DLC jump-selection method (Mathis et al., 2018; Gorssen et al., 2022) to capture windows of frames with high levels of uncertainty associated with significant movement. The second relies on DINOv2 high-dimensional embeddings (Oquab et al., 2024), allowing vector comparisons to maximize information gain by selecting frames with the largest distances from one another in the embedding space.

4.1.1. DISPLACEMENT SAMPLING

To capture the windows of maximum movement, inputs of N , the length of the full training set, M , the number of training frames, and W , the window size, are initialized to

$$N = \text{len}(\text{annotated video}), \quad M \in 50, 100, \quad W = 150. \quad (1)$$

The displacement algorithm segments the N frames into

windows of size W , creating n segments of size 150 with boundaries at frames

$$f_0, f_{150}, f_{300}, \dots, f_k, \quad (2)$$

where

$$k < N, \quad k \bmod 150 = 0 \quad (3)$$

The average Euclidean displacement of all keypoints in adjacent frame pairs

$$(f_i, f_{i+150}) \quad (4)$$

is then calculated as

$$d_{i,j} = d(f_i, f_j) = \frac{1}{P} \sum_{\ell=1}^P \sqrt{\left(f_{x_i}^{(\ell)} - f_{x_j}^{(\ell)}\right)^2 + \left(f_{y_i}^{(\ell)} - f_{y_j}^{(\ell)}\right)^2}, \quad (5)$$

where P is the number of keypoints and $f_{x_i}^{(\ell)}$ and $f_{y_i}^{(\ell)}$ denote the x - and y -coordinates of the ℓ th keypoint in frame i , respectively.

The displacement values are used to determine how many frames are sampled from each window; windows with greater total displacement are assigned a larger proportion of the available frames. Frames within each window are sampled randomly using a fixed seed of 42 to ensure reproducibility.

4.1.2. DINOv2 SAMPLING

Images can be represented as embeddings or vectors through various means, and recently large learning models, such as DINOv2 (Oquab et al., 2024), have proven successful at embedding images in an information-dense high-dimensional feature space (Oquab et al., 2024). Without the need for additional information such as labels, we input a given camera stream’s frames into the DINO vision transformer and receive a collection of vectors or embeddings that represents each image in the stack. These encodings are normalized and input into a k -center greedy algorithm, which initializes with a randomly selected frame f_0 . At each subsequent iteration, the next frame is selected to maximize its minimum distance to the set of frames already selected. Specifically, the next frame is chosen according to

$$f_i = \arg \max_{f \in F \setminus S} \min_{s \in S} d(f, s),$$

where S is the set of previously selected frames. This process is repeated until M frames have been selected, encouraging the selected frames to provide broad coverage of the embedding space (Gonzalez, 1985).

4.2. Model Training

After the training data were selected from the full set of labeled frames, we used a modified toolkit to convert X-ray video annotation to a DLC format (Laurence-Chasen et al., 2020). For machine learning development, we designed two experiments. The first experiment, called the “large-input study,” was used to describe the upper limit on the volume of input data and the number of training epochs. The second experiment, called the “model refinement study,” aimed to determine the minimum range of input data using diversity-based methods for data selection.

Both experiments used a pretrained Residual Neural Network 50 (ResNet-50) backbone with a heatmap Gaussian generator and a location refinement head trained for point tracking tasks, primarily learned on visible-light images. In every training instance, 95% of the input data was used for training and 5% for validation, along with the Adam optimizer and a scheduled learning rate from 0.0001 to 0.00001 over the course of the epochs. Training runs used a standard root mean squared error (RMSE) loss.

In the large-input study, we tested the modeling capability with two birds. Bird 1 had one trial and Bird 2 had two trials. For all videos, we sampled 200, 400, 800, and 1400 frames using the built-in random selection and data formatting method of DeepLabCut (Mathis et al., 2018). This method pulls frames at random based on the specified number of frames for training. We trained over 200 epochs and saved model snapshots every 25 epochs starting at epoch 100 to assess the evolution of the learning model. We implemented no additional augmentation except those built into DeepLabCut (Mathis et al., 2018). We then predicted on the left-out frames of the video and across all frames on the other trial.

In the model refinement study, we tested minimal model inputs using three trial videos of each Pig 1, Bird 2, and Bird 3. We first sampled 300 frames at random from each video and set these aside for testing. Then, for each of the three methods of training-data sampling described above—random, displacement (Mathis et al., 2018; Gorssen et al., 2022), and DINOv2 (Oquab et al., 2024) centers—100 and 50 frames were selected and input into the model. We trained the models for 125 epochs (Table 1). These quantities were informed by the results of the large-input study.

4.3. Location Refinement

In order to further reduce error after model inference, we implemented a location refinement algorithm as a post-processing step leveraging a difference-of-Gaussian blob detection algorithm implemented in scikit-image (Lowe, 2004) and the Hungarian algorithm (Kuhn, 1955) to link our predictions to the nearest blob. By passing two Gaussian

Table 1. Breakdown of models built for each animal and trial combination. Single-trial and joint-trial training sets were sampled using three different methods and two different quantities of training frames, resulting in 108 models.

Species	Name	Input Trials	Methods	Frames
Canary	Bird 2	15, 16, 17, 15 + 16, 15 + 17, 16 + 17	DINOV2, Displacement, Random	50, 100
Canary	Bird 3	5, 6, 10, 5 + 6, 5 + 10, 6 + 10	DINOV2, Displacement, Random	50, 100
Pig	Pig 1	1, 4, 6, 1 + 4, 1 + 6, 4 + 6	DINOV2, Displacement, Random	50, 100

filters of different widths, the computer can identify places of significant differences, including borders and dark/light spots within the Gaussian widths.

To ensure that anomalies, artifacts, and camera settings are accounted for in large-scale processing, we manually tuned the parameters σ_{ratio} , threshold, and overlap for one reference image from the video. We can match each prediction from the neural network to that of the blob detection using pairwise distances and then use the linear sum assignment (Hungarian) algorithm to calculate the minimal total distance of the pairings.

Let $\mathcal{P} = p_1, \dots, p_m$ denote the predicted keypoint locations and $\mathcal{B} = b_1, \dots, b_n$ denote the detected blob locations. Pairwise distances are calculated as

$$D_{ij} = \|p_i - b_j\|_2. \quad (6)$$

The assignment is then selected to minimize the total matching distance,

$$\pi^* = \arg \min_{\pi} \sum_i D_{i, \pi(i)}. \quad (7)$$

Since blobs are generally less than 3 pixels in diameter, a final 3×3 -pixel window around the prediction is used to select the pixel with the minimum value. For disk-storage optimization of this algorithm, we implement threading when running.

4.4. Evaluation Metrics

To understand the efficacy of our model, we tested for both accuracy and precision. As in Laurence-Chasen et al. (2020), pixel root mean squared error (RMSE) was used to measure

overall accuracy of the model. As corrections needed to be made, a metric that would indicate the number of correct labels per frame would be useful for those who had to annotate and indicate the precision of the model. The average Percentage of Labeled Points within 3 pixels (PP3) was used to capture the estimated number of points one would not have to correct.

For predicted keypoint coordinates $\hat{\mathbf{p}}_i$ and ground-truth coordinates $\mathbf{p}_i$, pixel RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2^2}. \quad (8)$$

PP3 is defined as the percentage of predicted keypoints whose Euclidean distance from the corresponding ground-truth keypoint is no greater than 3 pixels:

$$\text{PP3} = \frac{100}{N} \sum_{i=1}^N \mathbb{I}[\|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2 \leq 3]. \quad (9)$$

We test these across the dimensions of the training set, including input frame count and selection technique, multiple or singular videos as input, and whether the test set comes from the same video as training or an unseen video.

4.5. Baselines

All models made are compared to the hand-labeled points (pre-training) and manually corrected points (post-initial training). The models that were built and tested at different epochs are compared to 200-epoch models as the baseline. The models that were built and tested with different input-data selection methods (DINOV2 and Displacement) are tested against random selection as the baseline. The additional location refinement steps using difference-of-Gaussian detection are compared against the same model without additional location refinement steps.

5. Results

5.1. Overview

The experiments set out to assess the tunability and transferability of the DeepLabCut model using various training-data generation and sampling techniques and to consolidate these methods into a user-friendly, reproducible framework. The initial experiments, focusing on the quantity of input frames and number of training iterations, found that there was little improvement in model performance when inputting more than 400 annotated frames for model training (Figure 8). Increasing the number of input frames improved the prediction likelihood scores but left the pixel RMSE score unaffected.

In addition, the training and testing loss curves began to converge after 125 epochs (Figure 3).

Sampling training data from DINOv2 had a lower average pixel RMSE by 0.6 pixels and a higher average PP3 score by 1.2% relative to the random baseline across all sampled subjects; however, neither difference was statistically significant. PP3 was greatly improved by the addition of the blob-detection location-refinement algorithm, improving scores by 7.8% (95% CI: [6.70, 8.73], $t(359) = -14.955$, $p < 10^{-4}$) across all models trained and tested.

The graphical user interface (Figure 11) integrated the data annotation, machine learning training and inference, and data storage processes into a simple and informative tool. We reduced annotation time from 20 h to less than 4 h for 1500 frames with 20 radio-opaque markers and provided the user with a standardized data format ready for downstream biomechanics analysis.

5.2. The Large Input Study

We evaluated training time and model efficacy using 200, 400, 800, and 1400 frames randomly sampled from three videos, each containing approximately 2000 frames. Training time increased approximately linearly with the number of input frames, at approximately 35 minutes per 100 frames over 200 epochs. Models trained on a subset of frames from a trial video and evaluated on the remaining frames from the same video achieved a pixel RMSE of 0.86 ± 0.18 and a likelihood score of 0.96 ± 0.004 . In contrast, models trained on frames from one video and evaluated on a separate video achieved a pixel RMSE of 16.0 ± 16.3 and a likelihood score of 0.79 ± 0.04 (Figure 8).

5.3. Diversity Maximization for Input Data

To explore the effectiveness of each sampling method, we trained a total of 108 models (Table 1). Each of the three sampling methods was used to train 36 models, corresponding to two input frame counts across 18 animal-trial combinations. This experimental design allowed us to compare the robustness of the pipeline across different sampling methods and input sizes (Figure 4, 9).

Model training time averaged 10 ± 4 minutes for 50 input frames and 20 ± 11 minutes for 100 input frames, both trained for 150 epochs. Across sampling methods, mean training time was 15 ± 9 minutes for DINOv2, 15 ± 10 minutes for displacement, and 15 ± 11 minutes for random sampling.

Across all models, DINOv2-selected training data resulted in an average RMSE of 18.88 ± 24.35 pixels, compared with 18.98 ± 24.13 pixels for displacement sampling and 19.52 ± 23.72 pixels for random sampling. DINOv2 also produced the highest mean PP3 score at $81.07 \pm 19.39\%$,

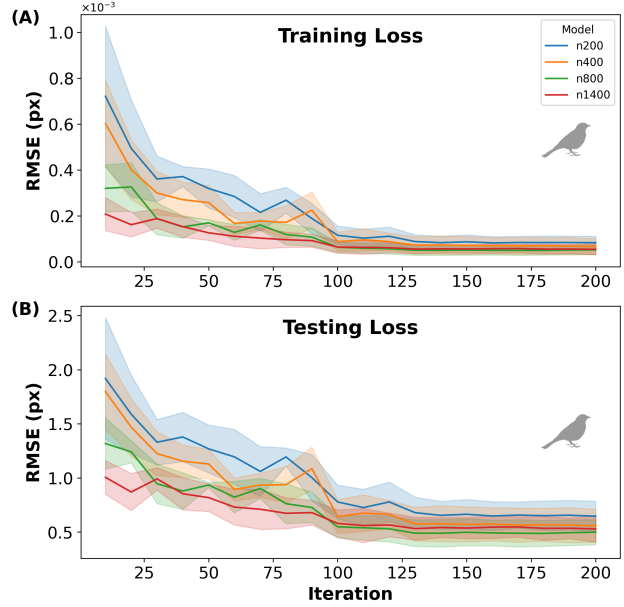


Figure 3. Training and evaluation dynamics for bird keypoint localization. Mean RMSE across the three trials is reported at increments of 25 iterations (epochs) for training (A) and testing (B). Shaded regions indicate ± 1 standard deviation across runs, with lower RMSE corresponding to more accurate keypoint localization. The bird silhouette illustrates the target animal represented in the dataset.

compared with $79.87 \pm 19.94\%$ for displacement sampling and $79.90 \pm 19.75\%$ for random sampling (Table 2).

When models were evaluated on unseen trials, DINOv2, displacement, and random sampling resulted in RMSEs of 32.08 ± 26.74 , 31.98 ± 26.52 , and 33.32 ± 25.14 pixels, respectively. The corresponding PP3 values were $67.19 \pm 16.88\%$, $65.78 \pm 16.78\%$, and $65.71 \pm 16.61\%$. For models evaluated on left-out data from trials represented in the training data, RMSEs were 3.79 ± 5.34 pixels for DINOv2, 4.11 ± 5.69 pixels for displacement, and 3.74 ± 4.33 pixels for random sampling. The corresponding PP3 values were $96.94 \pm 2.44\%$, $95.98 \pm 6.49\%$, and $96.12 \pm 5.08\%$, respectively (Figure 5).

When multiple trials were included in the training dataset, the average model RMSE was 8.81 ± 13.82 pixels for DINOv2 sampling, 7.85 ± 10.28 pixels for displacement sampling, and 8.22 ± 11.02 pixels for random sampling. For models trained using only a single trial, RMSE was 28.95 ± 28.25 , 30.10 ± 28.59 , and 30.81 ± 27.46 pixels for DINOv2, displacement, and random sampling, respectively (Figure 5).

Across animal types, DINOv2 sampling resulted in an RMSE of 15.27 ± 25.55 pixels for birds and 24.29 ± 21.55 pixels for pigs. For comparison, displacement sampling resulted in RMSEs of 18.22 ± 27.84 pixels for birds and

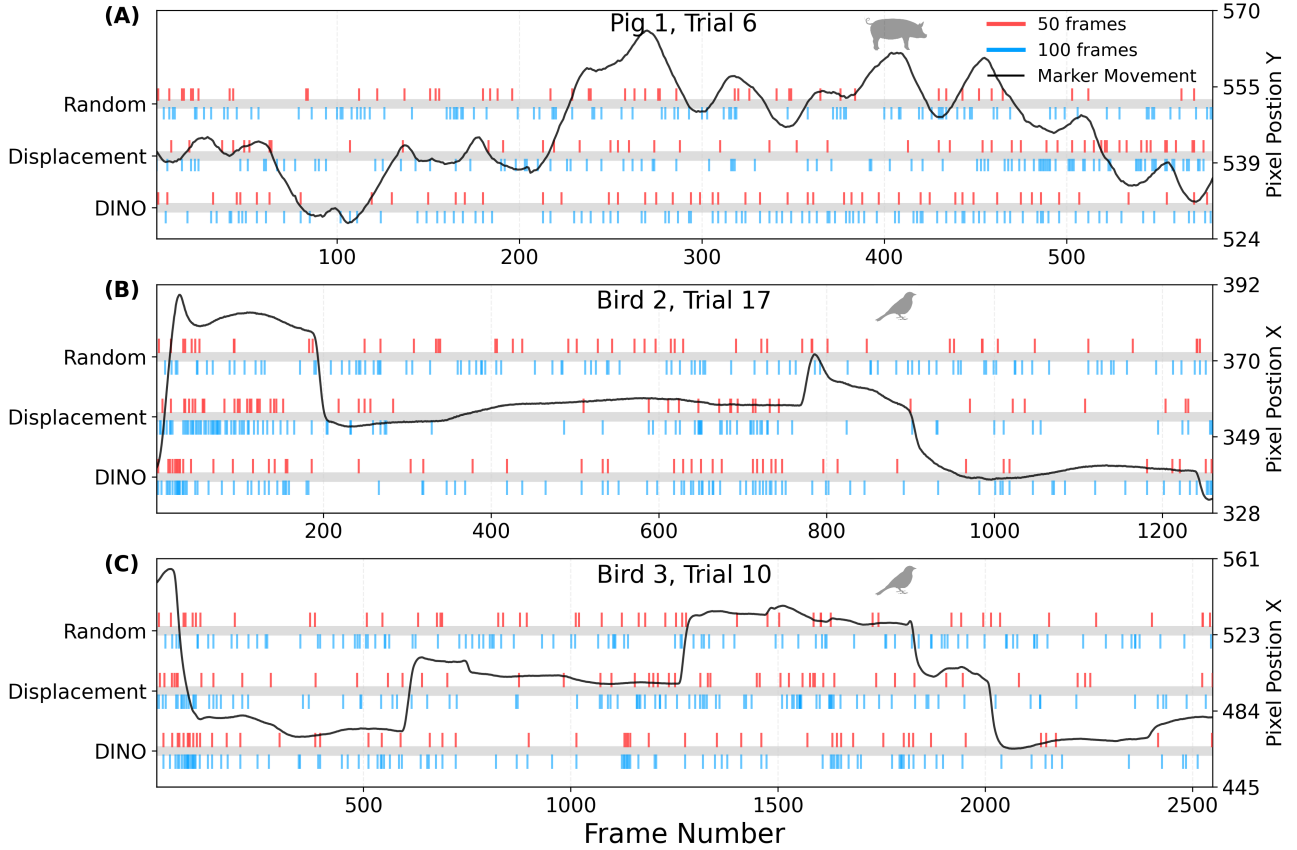


Figure 4. Frame selection strategies across representative animal trials. Selected frames are shown across three representative trials from Pig 1 (A, top), Bird 2 (B, middle), and Bird 3 (C, bottom). For each sampling method (Random, Displacement, and DINOv2), 50 and 100 frames were selected for training. Colored vertical marks indicate selected frames, while the black trajectory shows the corresponding mean keypoint position over time. The figure illustrates how sampling strategies distribute training frames across trials with differing patterns of animal movement.

20.10 $\pm$ 17.37 pixels for pigs, while random sampling resulted in 17.93 $\pm$ 25.98 pixels and 21.89 $\pm$ 19.89 pixels, respectively. The corresponding PP3 values for DINOv2 were 78.51 $\pm$ 22.06% for birds and 84.92 $\pm$ 13.83% for pigs (Table 2).

With 50 training frames, PP3 was 78.90 $\pm$ 20.22% for DINOv2 sampling, 77.92 $\pm$ 20.33% for displacement sampling, and 78.82 $\pm$ 19.39% for random sampling. The corresponding pixel RMSE values were 21.47 $\pm$ 27.52, 19.55 $\pm$ 23.73, and 20.07 $\pm$ 23.35 pixels, respectively. With 100 training frames, PP3 increased to 83.25 $\pm$ 18.43% for DINOv2, compared with 81.82 $\pm$ 19.51% for displacement and 80.99 $\pm$ 20.21% for random sampling. Pixel RMSE was 16.29 $\pm$ 20.60, 18.40 $\pm$ 24.71, and 18.96 $\pm$ 24.28 pixels for DINOv2, displacement, and random sampling, respectively (Table 2).

5.4. Location Refinement

The blob-detection and alignment location-refinement step substantially improved PP3 while leaving pixel RMSE largely unchanged (Figure ??). Across all models, PP3 increased from 80.3 $\pm$ 19.7% at baseline to 88.0 $\pm$ 12.57% following refinement, corresponding to an improvement of approximately 7.8 percentage points. Pixel RMSE remained similar, at approximately 19.1 pixels after refinement compared with 19.3 pixels at baseline (Figure 7).

Across both species, models evaluated on unseen trials showed an increase in PP3 from 66.23 $\pm$ 16.68% to 79.49 $\pm$ 11.79% following the refinement step ($p = 3.45 \times 10^{-44}$). Pixel RMSE remained effectively unchanged, with values of 32.46 $\pm$ 26.02 pixels before refinement and 32.53 $\pm$ 26.01 pixels after refinement ($p = 8.61 \times 10^{-5}$) (Table 3).

For birds, PP3 increased from 77.45 $\pm$ 22.39% to 88.10 $\pm$ 12.08% following refinement ($p = 2.98 \times 10^{-30}$). For pigs, PP3 increased from 84.53 $\pm$ 13.61% to 87.84 $\pm$ 13.31% ($p = 1.93 \times 10^{-45}$). The corresponding pixel RMSE values

Table 2. Baseline keypoint localization performance for models trained on data from each sampling method: DINOv2, Displacement, and Random. Each grouping shows the variation in results across input parameters. Pair Type “Across” indicates that the training and testing trials differed, while “Within” indicates that training and testing data were from the same trial. A true combined model indicates that the model was trained on data from more than one trial. Values are reported as mean $\pm$ standard deviation. Best-performing values within each metric and row are shown in bold.

Grouping	Condition	PP3 (%)			RMSE (px)		
		DINOv2	Displacement	Random	DINOv2	Displacement	Random
Overall	–	80.28 $\pm$ 19.65	79.87 $\pm$ 19.94	79.90 $\pm$ 19.75	19.12 $\pm$ 24.00	18.98 $\pm$ 24.13	19.52 $\pm$ 23.72
Pair Type	Across	67.19 $\pm$ 16.88	65.78 $\pm$ 16.78	65.71 $\pm$ 16.61	32.08 $\pm$ 26.74	31.98 $\pm$ 26.52	33.32 $\pm$ 25.14
Pair Type	Within	96.94 $\pm$ 2.44	95.98 $\pm$ 6.49	96.12 $\pm$ 5.08	3.79 $\pm$ 5.34	4.11 $\pm$ 5.69	3.74 $\pm$ 4.33
Animal	Bird	78.51 $\pm$ 22.06	76.95 $\pm$ 22.86	76.90 $\pm$ 22.51	15.27 $\pm$ 25.55	18.22 $\pm$ 27.84	17.93 $\pm$ 25.98
Animal	Pig	84.92 $\pm$ 13.83	84.26 $\pm$ 13.57	84.42 $\pm$ 13.70	24.29 $\pm$ 21.55	20.10 $\pm$ 17.37	21.89 $\pm$ 19.89
Training Frames	50	78.90 $\pm$ 20.22	77.92 $\pm$ 20.33	78.82 $\pm$ 19.39	21.47 $\pm$ 27.52	19.55 $\pm$ 23.73	20.07 $\pm$ 23.35
Training Frames	100	83.25 $\pm$ 18.43	81.82 $\pm$ 19.51	80.99 $\pm$ 20.21	16.29 $\pm$ 20.60	18.40 $\pm$ 24.71	18.96 $\pm$ 24.28
Combined Model	False	72.72 $\pm$ 22.11	70.55 $\pm$ 22.41	70.77 $\pm$ 22.12	28.95 $\pm$ 28.25	30.10 $\pm$ 28.59	30.81 $\pm$ 27.46
Combined Model	True	89.42 $\pm$ 11.30	89.19 $\pm$ 11.08	89.04 $\pm$ 11.30	8.81 $\pm$ 13.82	7.85 $\pm$ 10.28	8.22 $\pm$ 11.02

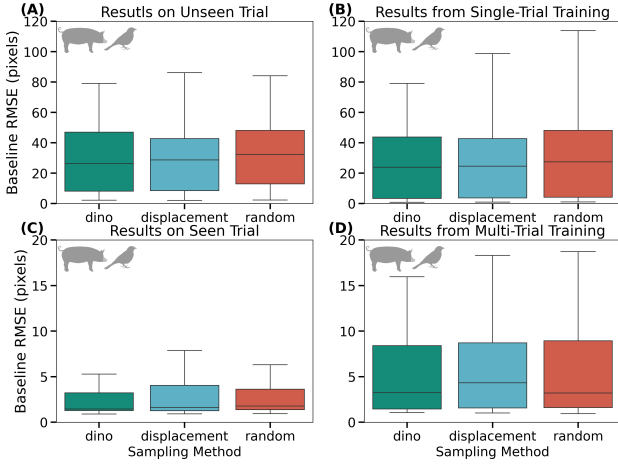


Figure 5. Baseline keypoint localization error across sampling strategies and evaluation conditions. Results are shown for models trained on one or more trials and evaluated on (A) an unseen trial, (B) models trained without trial combination, (C) within-animal comparisons, and (D) combined models. Boxplots show the distribution of baseline RMSE (pixels) for Random, Displacement, and DINOv2 sampling strategies; lower RMSE indicates more accurate keypoint localization. Results are shown separately for bird and pig datasets.

for birds were 17.14 ± 26.39 pixels before refinement and 17.42 ± 26.24 pixels after refinement. For pigs, RMSE was 22.10 ± 19.62 pixels before refinement and 22.13 ± 19.69 pixels after refinement (Figure 7, 10).

The effect of refinement was also observed across training-set sizes. At baseline, the difference in PP3 between models trained with 50 and 100 frames was approximately 3.4 percentage points, with scores of 78.6% and 82.0%, respectively. Following location refinement, this difference decreased to approximately 1.3 percentage points, with PP3 scores of 87.4% for 50 frames and 88.6% for 100 frames (Table 3).

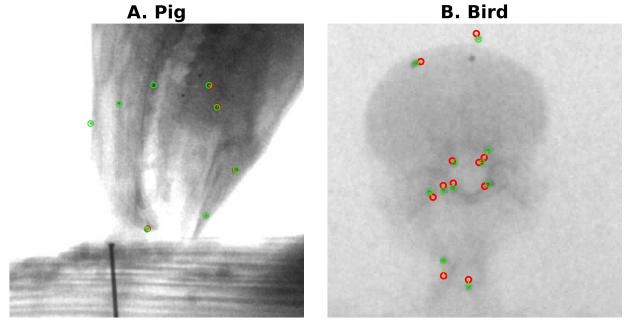


Figure 6. Example of blob-based keypoint correction for (A) Pig 1 and (B) Bird 2 images. Red circles indicate the original model predictions, while green circles indicate predictions corrected using detected image blobs. Yellow dashed lines connect the original and corrected locations, illustrating cases where predictions are displaced to the center of a nearby detected blob. Notice (middle of pig mouth and top of Bird head) predictions far outside the range of the truth will not be corrected accurately. Blob detection parameters are adapted to the animal type to account for differences in marker appearance.

6. Discussion & Conclusion:

6.1. Key Findings

6.1.1. LARGE INPUT STUDY, DATA DIVERSITY

We found that increasing sample size or training iterations was not a major driver of model success. That is, accuracy and precision of the trained algorithm is more influenced by the type and kind of data that is input at the model refinement stage. We were directed towards a workflow that capitalized on data diversity (that is, how different images are from one-another) for image selection. Using 50 annotated frames from one trial (selected by DINOv2), the user can train a model that predicts 67% of points to within 3 pixels of accuracy but an average RMSE of 32.08 pixels on other trials of the same animal. This gives the user initial labels

Table 3. Blob-detection-refined keypoint localization performance for models trained on data from each sampling method: DINOv2, Displacement, and Random. Each grouping shows the variation in results across input parameters. Pair Type “Across” indicates that the training and testing trials differed, while “Within” indicates that training and testing data were from the same trial. A true combined model indicates that the model was trained on data from more than one trial. Values are reported as mean $\pm$ standard deviation. Best-performing values within each metric and row are shown in bold.

Grouping	Condition	Percent Points within 3 pixels (PP3) (%)			RMSE (px)		
		DINO	Displacement	Random	DINO	Displacement	Random
Overall	–	88.42 $\pm$ 12.79	87.78 $\pm$ 12.61	87.78 $\pm$ 12.40	19.06 $\pm$ 24.26	19.16 $\pm$ 24.05	19.68 $\pm$ 23.63
Pair Type	Across	80.12 $\pm$ 12.56	79.19 $\pm$ 11.68	79.16 $\pm$ 11.26	32.15 $\pm$ 26.75	32.07 $\pm$ 26.52	33.39 $\pm$ 25.11
Pair Type	Within	97.91 $\pm$ 1.11	97.61 $\pm$ 1.84	97.64 $\pm$ 1.40	4.10 $\pm$ 5.20	4.41 $\pm$ 5.52	4.01 $\pm$ 4.19
Animal	Bird	88.66 $\pm$ 12.30	87.82 $\pm$ 12.22	87.81 $\pm$ 11.88	15.57 $\pm$ 25.43	18.50 $\pm$ 27.68	18.19 $\pm$ 25.82
Animal	Pig	88.06 $\pm$ 13.63	87.73 $\pm$ 13.30	87.74 $\pm$ 13.27	24.30 $\pm$ 21.62	20.16 $\pm$ 17.47	21.92 $\pm$ 19.96
Training Frames	50	87.27 $\pm$ 14.02	87.29 $\pm$ 12.64	87.49 $\pm$ 12.45	21.61 $\pm$ 27.48	19.69 $\pm$ 23.68	20.19 $\pm$ 23.27
Training Frames	100	89.58 $\pm$ 11.43	88.28 $\pm$ 12.66	88.07 $\pm$ 12.45	16.51 $\pm$ 20.48	18.63 $\pm$ 24.60	19.17 $\pm$ 24.17

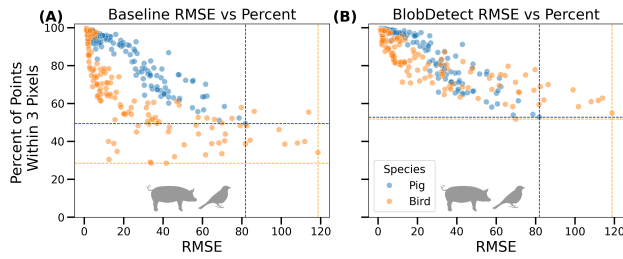


Figure 7. Relationship between localization error and keypoint accuracy. (A) Baseline pixel RMSE versus the percentage of keypoints localized within 3 pixels of the ground truth and (B) the corresponding relationship after blob-detection-based location refinement. Each point represents an evaluation result, with colors indicating animal species. Dashed lines indicate the minimum accuracy and maximum RMSE observed for each species.

(i.e. points) which the user can then correct. We found that by again selecting and correcting 50 out of the predicted frames with DINOv2, the user can re-train and re-predict on the same video and get 97% of points to within 3 pixels and an RMSE score of 3.8 pixels. This process takes roughly 50 minutes for correcting, 10 minutes for retraining and 3 minutes for repredicting. We found that this quick, iterative process (see Figure 2, steps 1, 4A, and 4B) is what is more effective for model accuracy and precision.

6.1.2. LOCATION REFINEMENT

Furthermore, we found the addition of a new location refinement method as a post-processing step after model prediction shows an increase of $\sim 7\%$ PP3 on average across all experiments run when compared to non location-refined data. Specifically, when predicting on a new video (Step 3b), an increase in PP3 by 15% can be seen. This provides a starting point of 79% points predicted within 3 pixels of the truth (hand-tracked points) from the aforementioned 64%. Then, within trial PP3 increases from 97% to 98%.

6.2. Context

This work shows how adaptation of existing large foundational models for specific tasks can be optimized with hyperparameter tuning and pre/post-processing algorithms into a single data-processing pipeline that allows models to be applied to, and achieve success in, broad real world applications (Rolnick et al., 2025). After previous works showed the efficacy of DLC for X-ray video annotation (Laurence-Chasen et al., 2020; Kirkpatrick et al., 2022; Andries et al., 2025; Flotho et al., 2024), we improved efforts to bring machine learning to the hands of biomechanics researchers. Key to this accessibility is our construction of a graphical user interface (GUI) that leads the user through the steps of tuning and retraining while retaining the layout of the existing labeling software XMALab, promoting adoption within the scientific community. This iterative, deep learning-assisted, approach allows for interaction with the data at each step in the process, which prompts the user to simultaneously build hypotheses and improve the models predictive power over time through retraining and combined training while remaining time and resource efficient.

6.3. Strengths and limitations

While we tested our modeling pipeline with various hyperparameter adjustments to minimize cost (e.g computational power and processing time) and maximize usability, our datasets were limited to a sample size of 9 unique videos. Our pipeline successfully handles a range of both marker quantities and movements. However, analyzing the entire frame leaves room for artifacts that may challenge the model’s predictive ability. The data set itself also contains dimensions that we did not utilize: time and a third spatial dimension (i.e. the reconstructed dimension from biplanar X-ray data). The lack of time-dependence for the model limits its ability to self-correct in the same way a user corrects the data, that is, based on neighboring frames which may have more clear points to track without much frame to frame

displacement (Girdhar et al., 2018). Works by DeepLabCut (Mathis et al., 2018) and XMA Lab (Brainerd et al., 2010; Huffman et al., 2016; Knörlein et al., 2016) have incorporated a calibration image to capture the three-dimensional nature of the videos being processed, yet ours works directly with the two dimensional nature of the frames and relies on the user to reconstruct the data in three dimensions after processing.

6.4. Future Directions

In the future, we plan to test the feasibility of incorporating time-dependence and three-dimensional information into the model-training pipeline, as well as refinement of pre- and post-training steps for efficiency and ease of use. Currently, the hyperparameters tuned for location refinement blob detection are done by hand, yet future work could set up a simple learning pipeline of the key parameters through simple least squares regression or other (Bergstra et al., 2013; Efe et al., 2021). In addition, existing or custom object detection and segmentation models, such as YOLO (Redmon et al., 2016) or SAM 2 (Ravi et al., 2025), could be used to constrain predictions to the region occupied by the animal. Blob detection and region prediction could even be applied to new trials in the future, assisting with the creation of training datasets. Furthermore, the ResNet-50 model (He et al., 2015) has millions of parameters and was built to be a generalist, yet our question investigates a smaller domain and specific task, so informed pruning could be done to reduce training and inference time and computational resources.

This pipeline could also be augmented to extract information previously untrackable by the researcher. For example, in the canary datasets, the heart and body shape are visible in most of the frames. Object detection and segmentation models, such as the aforementioned YOLO (Redmon et al., 2016), SAM 2 (Ravi et al., 2025), or nninteractive (Isensee et al., 2024), could extract physiological information such as heart rate and breathing rate even as it uses that information to inform marker-tracking. More work is needed to test if these two types of datasets are mutually informative or if separate pipelines are needed.

6.5. Conclusion

Our work confirms that the advent of artificial intelligence should not act as a replacement of Science or researchers, but rather as an amplification device. Using active learning, we built a pipeline that significantly shortens data digitization time while retaining accuracy and precision. With each application of these complex deep learning algorithms, we are moving towards models being more accessible, more resource efficient, and more productive so research can unveil both large and small scientific discoveries that were

formerly unreachable when examined by hand or simple algorithms.

Data & Code

The miniature pig XROMM dataset is publicly available through the XMAPortal: https://xmaportal.org/webportal/larequest.php?request=studyOverview_public\&StudyID=11\&instit=BROWN

The code for this study can be found at: <https://github.com/tcmarino8/BirdSongs-MNHN>

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A. Supplemental

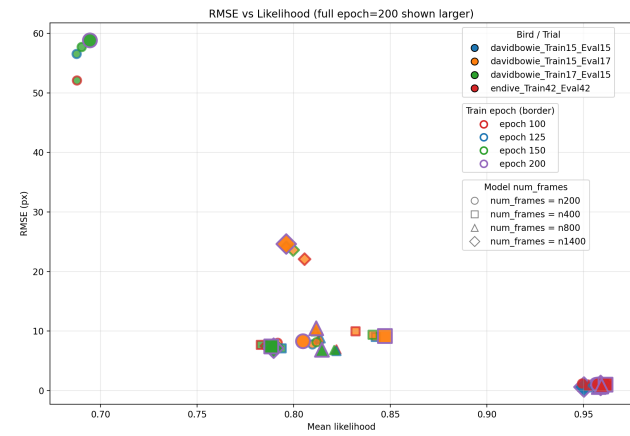


Figure 8. The comparison of RMSE vs Likelihood for all models tested on data in the Large input study

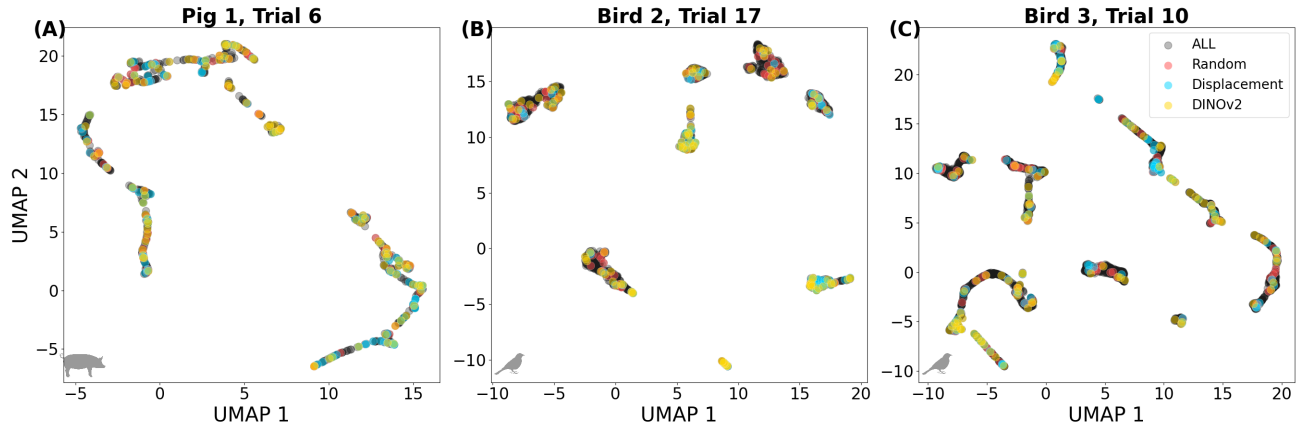


Figure 9. Training-frame selection across the embedded data distribution. (A) Pig 1, (B) Bird 2, and (C) Bird 3 trials. Gray points represent all frames in the respective trial, while colored points indicate frames selected for training by Random, Displacement, and DINO sampling. The UMAP embeddings provide a qualitative comparison of how each strategy covers the visual diversity of the underlying video data.

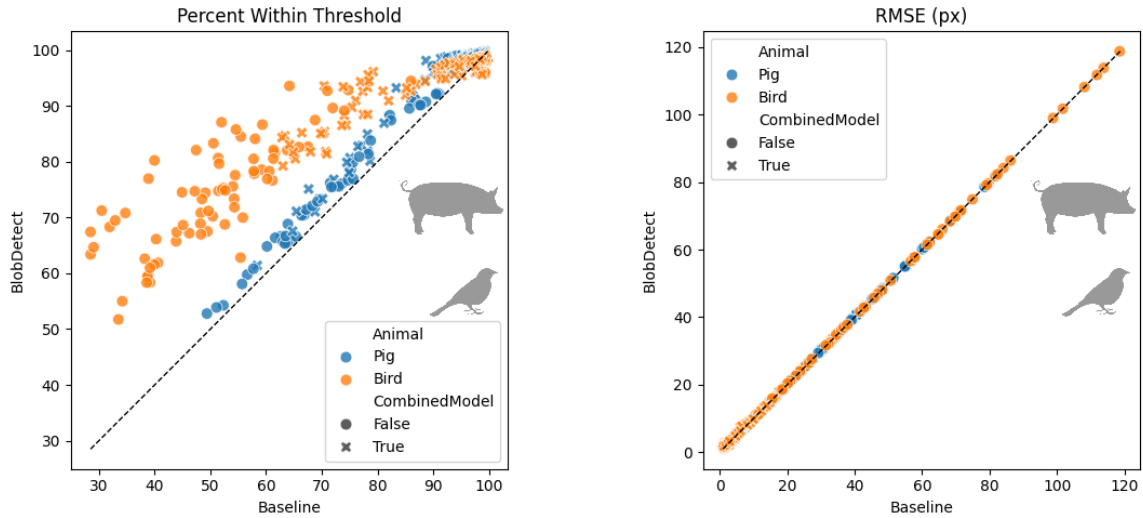


Figure 10. The comparison between PP3 (left) and RMSE (right) scores between baseline (x) and blob detect (y). Dashed line is to show $y = x$. Point color is animal, and point shape indicates if more than one trial went into the training set.

Table 4. Keypoint localization performance before and after blob-detection-based location refinement across sampling methods, trial pairing, animal type, and training-set size. Values are reported as mean $\pm$ standard deviation. N denotes the number of evaluations.

Grouping	Method	Condition	N	PP3 (%)		RMSE (px)	
				Baseline	Refined	Baseline	Refined
Overall	–	–	360	80.28 $\pm$ 19.65	88.00 $\pm$ 12.57	19.12 $\pm$ 24.00	19.30 $\pm$ 23.92
Sampling Method							
	DINOv2	–	120	81.07 $\pm$ 19.39	88.42 $\pm$ 12.79	18.88 $\pm$ 24.35	19.06 $\pm$ 24.26
	Displacement	–	120	79.87 $\pm$ 19.94	87.78 $\pm$ 12.61	18.98 $\pm$ 24.13	19.16 $\pm$ 24.05
	Random	–	120	79.90 $\pm$ 19.75	87.78 $\pm$ 12.40	19.52 $\pm$ 23.72	19.68 $\pm$ 23.63
Pair Type							
	–	Across	192	66.23 $\pm$ 16.68	79.49 $\pm$ 11.79	32.46 $\pm$ 26.02	32.53 $\pm$ 26.01
	–	Within	168	96.35 $\pm$ 4.95	97.72 $\pm$ 1.48	3.88 $\pm$ 5.13	4.18 $\pm$ 4.98
Animal							
	–	Bird	216	77.45 $\pm$ 22.39	88.10 $\pm$ 12.08	17.14 $\pm$ 26.39	17.42 $\pm$ 26.24
	–	Pig	144	84.53 $\pm$ 13.61	87.84 $\pm$ 13.31	22.10 $\pm$ 19.62	22.13 $\pm$ 19.69
Training Frames							
	–	50 frames	180	78.55 $\pm$ 19.88	87.35 $\pm$ 12.98	20.36 $\pm$ 24.81	20.50 $\pm$ 24.76
	–	100 frames	180	82.02 $\pm$ 19.31	88.64 $\pm$ 12.14	17.89 $\pm$ 23.17	18.10 $\pm$ 23.06
Combined Model							
	–	False	180	71.35 $\pm$ 22.11	82.14 $\pm$ 13.93	29.96 $\pm$ 27.96	30.09 $\pm$ 27.87
	–	True	180	89.22 $\pm$ 11.16	93.85 $\pm$ 7.35	8.29 $\pm$ 11.75	8.51 $\pm$ 11.68
Sampling Method + Pair Type							
	DINOv2	Across	64	67.19 $\pm$ 16.88	80.12 $\pm$ 12.56	32.08 $\pm$ 26.74	32.15 $\pm$ 26.75
	DINOv2	Within	56	96.94 $\pm$ 2.44	97.91 $\pm$ 1.11	3.79 $\pm$ 5.34	4.10 $\pm$ 5.20
	Displacement	Across	64	65.78 $\pm$ 16.78	79.19 $\pm$ 11.68	31.98 $\pm$ 26.52	32.07 $\pm$ 26.52
	Displacement	Within	56	95.98 $\pm$ 6.49	97.61 $\pm$ 1.84	4.11 $\pm$ 5.69	4.41 $\pm$ 5.52
	Random	Across	64	65.71 $\pm$ 16.61	79.16 $\pm$ 11.26	33.32 $\pm$ 25.14	33.39 $\pm$ 25.11
	Random	Within	56	96.12 $\pm$ 5.08	97.64 $\pm$ 1.40	3.74 $\pm$ 4.33	4.01 $\pm$ 4.19
Sampling Method + Animal							
	DINOv2	Bird	72	78.51 $\pm$ 22.06	88.66 $\pm$ 12.30	15.27 $\pm$ 25.55	15.57 $\pm$ 25.43
	DINOv2	Pig	48	84.92 $\pm$ 13.83	88.06 $\pm$ 13.63	24.29 $\pm$ 21.55	24.30 $\pm$ 21.62
	Displacement	Bird	72	76.95 $\pm$ 22.86	87.82 $\pm$ 12.22	18.22 $\pm$ 27.84	18.50 $\pm$ 27.68
	Displacement	Pig	48	84.26 $\pm$ 13.57	87.73 $\pm$ 13.30	20.10 $\pm$ 17.37	20.16 $\pm$ 17.47
	Random	Bird	72	76.90 $\pm$ 22.51	87.81 $\pm$ 11.88	17.93 $\pm$ 25.98	18.19 $\pm$ 25.82
	Random	Pig	48	84.42 $\pm$ 13.70	87.74 $\pm$ 13.27	21.89 $\pm$ 19.89	21.92 $\pm$ 19.96
Sampling Method + Training Frames							
	DINOv2	50 frames	60	78.90 $\pm$ 20.22	87.27 $\pm$ 14.02	21.47 $\pm$ 27.52	21.61 $\pm$ 27.48
	DINOv2	100 frames	60	83.25 $\pm$ 18.43	89.58 $\pm$ 11.43	16.29 $\pm$ 20.60	16.51 $\pm$ 20.48
	Displacement	50 frames	60	77.92 $\pm$ 20.33	87.29 $\pm$ 12.64	19.55 $\pm$ 23.73	19.69 $\pm$ 23.68
	Displacement	100 frames	60	81.82 $\pm$ 19.51	88.28 $\pm$ 12.66	18.40 $\pm$ 24.71	18.63 $\pm$ 24.60
	Random	50 frames	60	78.82 $\pm$ 19.39	87.49 $\pm$ 12.45	20.07 $\pm$ 23.35	20.19 $\pm$ 23.27
	Random	100 frames	60	80.99 $\pm$ 20.21	88.07 $\pm$ 12.45	18.96 $\pm$ 24.28	19.17 $\pm$ 24.17

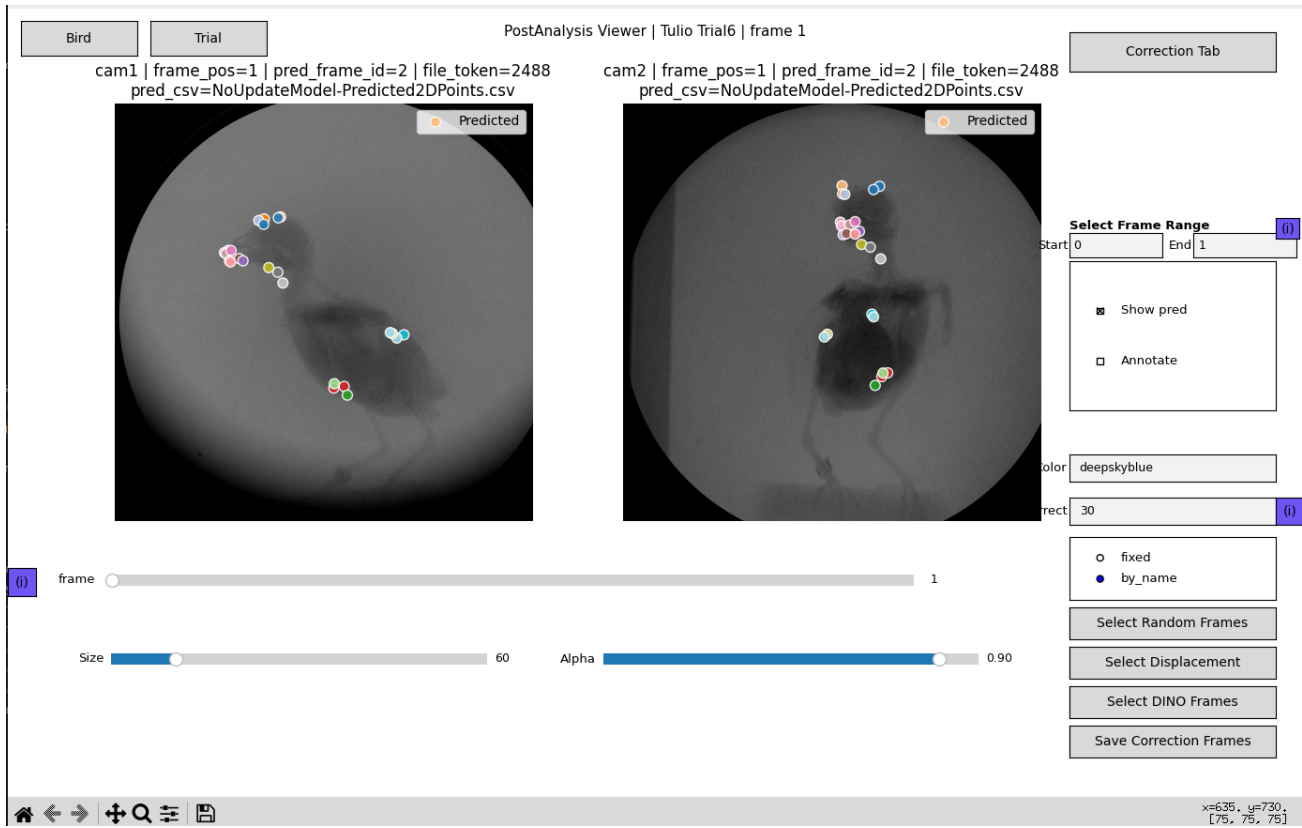


Figure 11. The Graphical User Interface (GUI) that integrates our workflow. The two frames shown are taken at the same time from each of the two cameras. Designed to give easy access to frame selection and model updating, buttons and sliders are accompanied by information popups.